

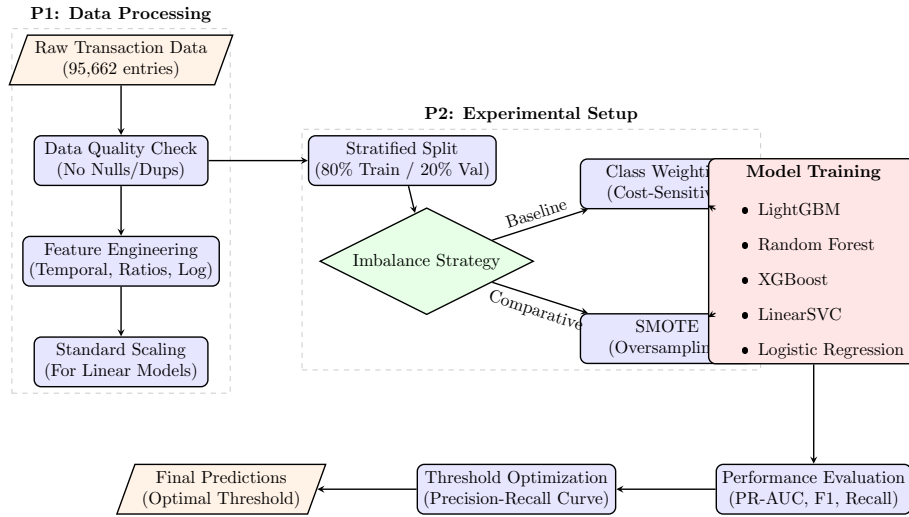


Figure 1: Methodological Workflow from Data Ingestion to Model Optimisation

The framework follows a CRISP-DM-aligned pipeline, incorporating parallel imbalance-handling strategies and threshold optimisation for fraud detection.